

Harry Myers CS m/46 HW #1 April 21st 2026

HW #1

1. a. $J(w_0, w_1) = \sum_n a_n (w_0 + w_1 x_{n,1} - y_n)^2$

$\frac{\partial J}{\partial w_0} = 2 \sum_n a_n (w_0 + w_1 x_{n,1} - y_n)$, $\frac{\partial J}{\partial w_1} = 2 \sum_n a_n x_{n,1} (w_0 + w_1 x_{n,1} - y_n)$

b. If one weight $a_n \rightarrow \infty$ that means point n is infinitely important to the line must go through k . This completely messes up the optimal solution.

Cont. R2: $J_1(w_0, w_1) = \sum_n a_n (w_0 + w_1 x_{n,1} - y_n)^2 + \lambda (w_0^2 + w_1^2)$

$\frac{\partial J_1}{\partial w_0} = 2 \sum_n a_n (w_0 + w_1 x_{n,1} - y_n) + 2 \lambda w_0$, $\frac{\partial J_1}{\partial w_1} = 2 \sum_n a_n x_{n,1} (w_0 + w_1 x_{n,1} - y_n) + 2 \lambda w_1$

1.C. II. normal equations $\nabla J = 0$ $\theta = (X^T X)^{-1} X^T Y$ comes from minimizing standard loss $J(w) = \frac{1}{2} \|Xw - y\|_2^2$

Add weights a_n $J(w) = \sum_n a_n (w_0 + w_1 x_{n,1} - y_n)^2$ $A = \text{diag}(a_n) = \begin{bmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{bmatrix}$

$Jw = (Xw - y)^T A (Xw - y)$ (matrix form) $+ \lambda \|w\|^2$

$J_1(w_0, w_1) = \sum_n a_n (w_0 + w_1 x_{n,1} - y_n)^2 + \lambda (w_0^2 + w_1^2)$

$X = \begin{bmatrix} 1 & x_{1,1} \\ 1 & x_{2,1} \\ \vdots & \vdots \\ 1 & x_{n,1} \end{bmatrix}$ $w = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$ $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$

$\frac{\partial J_1}{\partial w_0} = 2 \sum_n a_n (w_0 + w_1 x_{n,1} - y_n) + 2 \lambda w_0 = 0$ $\frac{\partial J_1}{\partial w_1} = 2 \sum_n a_n x_{n,1} (w_0 + w_1 x_{n,1} - y_n) + 2 \lambda w_1 = 0$

$\begin{bmatrix} 1 & A^T X \\ X^T A & X^T A X \end{bmatrix} w = \begin{bmatrix} X^T A y \\ X^T A X w - X^T A y + \lambda w = 0 \end{bmatrix}$ $X^T A y = w (X^T A X + \lambda I)$

$w^k = (X^T A X + \lambda I)^{-1} X^T A y$

2. logistic loss $h_w(x) = \sigma(w^T x) = \frac{1}{1 + e^{-w^T x}}$

a. when $|w^T x|$ is large $e^{-w^T x} \rightarrow 0$ $\times$ $h_w(x) \rightarrow 1$ or $e^{-w^T x} \rightarrow \infty$ $\times$ $h_w(x) \rightarrow 0$ so always predicts same

b. $h_w(x) = 0.5 = \sigma(w^T x)$ $\frac{1}{2}$ when $e^{-w^T x} = 1$ $\times$ $-w^T x$ is 0 $\times$ $w^T x$ is linear in x

c. $\nabla_w J_{ce} = (\sigma(w^T x) - y) x$ $\nabla_w J_{mse} = (\sigma(w^T x) - y) \sigma'(w^T x) x$ $\sigma'(w^T x) = \left(\frac{1}{1+e^{-x}}\right)' = \frac{e^x}{(e^x + 1)^2} = \frac{1}{(1+e^{-x})} \left(\frac{1+e^{-x}}{1+e^{-x}}\right) - 1$

$\frac{1}{1+e^{-x}} = 1 - \frac{1}{1+e^x} < \sigma'(z) (1 - \sigma(z)) = 1$ if $\sigma(w^T x) = 1$ then $\sigma'(w^T x) = 0$

else if $\sigma(w^T x) = 0$ then $\sigma'(w^T x) = 0$ cross entropy gradients: if a model is very wrong it learns fast but if the model is wrong on squared loss it won't learn very much.

d. cross entropy $\rightarrow$ squared loss only affects the training dynamics cause

both loss functions use same model $h(x) = \sigma(w^T x)$ & same threshold to produce some decision boundary

The loss function only affects how fast you get to w . once you have w the boundary looks the same.

3. a. $J(w) = \|xw - y\|_2^2 + \lambda \|w\|_2^2$ $x \in \mathbb{R}^{n \times d}$, $w \in \mathbb{R}^d$, $y \in \mathbb{R}^n$ $\lambda > 0$

$$J(w) = \left\| \begin{bmatrix} x \\ \sqrt{\lambda} I \end{bmatrix} w - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|_2^2 = \left\| \begin{bmatrix} xw \\ \sqrt{\lambda} w \end{bmatrix} - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|_2^2 = \|xw - y\|_2^2 + \|\sqrt{\lambda} w\|_2^2$$

$$Jw = \|xw - y\|_2^2 + \lambda \|w\|_2^2 = \|xw - y\|_2^2 + \|\sqrt{\lambda} w\|_2^2 = \left\| \begin{bmatrix} xw - y \\ \sqrt{\lambda} w \end{bmatrix} \right\|_2^2 = \left\| \begin{bmatrix} x \\ \sqrt{\lambda} I \end{bmatrix} w - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|_2^2$$

3. b. $\frac{\lambda}{\lambda + 1}$ 3 b. I. a data points are being added

$$\text{If each fake data point has only 1 feature } \tilde{x} = \begin{bmatrix} x \\ \sqrt{\lambda} I \end{bmatrix} \begin{bmatrix} y \\ 0 \end{bmatrix}$$

equal to $\sqrt{\lambda}$ & all others are zero & the y for this data point is zero this

means that whenever feature x is active the function reduces that weight. This is meant

to regularize the function & keep the weights smaller.